Gas Laws

21. (c) $V \propto T \Rightarrow \frac{V_1}{V_2} = \frac{T_1}{T_2} \Rightarrow \frac{V}{2V} = \frac{(273+27)}{T_2} = \frac{300}{T_2}$

$\Rightarrow T_2 = 600K = 327°C$

22. (b) The collision of molecules of ideal gas is elastic collision.

23. (a) In free expansion of Vander waal's gas, its temperature decreases

24. (a) According to the principle of homogeneity dimensional of P = Dimension of $\frac{a}{V^2}$

$\Rightarrow \frac{a}{[L^3]^2} = [ML^{-1}T^{-1}] \Rightarrow a = [ML^5L^{-2}]$

25. (a) According to Avogadro hypothesis

26. (a) $P \propto T \Rightarrow \frac{P_1}{P_2} = \frac{T_1}{T_2} \Rightarrow \frac{P_2 - P_1}{P_1} = \frac{T_2 - T_1}{T_2}$

$\Rightarrow \left(\frac{\Delta P}{P}\right)\% = \left(\frac{251-250}{250}\right) \times 100 = 0.4\ \%$

27. (c) Here temperature remain constant

so $P_1V_1 = P_2V_2 \Rightarrow 76 \times 5 = P_2 \times 35$

$\Rightarrow P_2 = \frac{76 \times 5}{35} = 10.85 cmofhg$

28. (a) $V \propto T$ (as constant pressure)

29. (c) At constant temperature ; PV = constant

$\Rightarrow P \times \left(\frac{m}{D}\right)$ = constant

$\Rightarrow \frac{P}{D}$ = constant = K. [D = Density]

30. (c) Boyle's and Charle's law follows kinetic theory of gases.

31. (d) $P \propto T \Rightarrow \frac{P_2}{P_1} = \frac{T_2}{T_1} = \frac{(273+100)}{(273+0)} = \frac{373}{273} \Rightarrow P_2 = \frac{760 \times 373}{273} = 1038mm$

32. (c) $V \propto T \Rightarrow \frac{V_1}{V_2} = \frac{T_1}{T_2}$

$\Rightarrow \frac{V}{V_2} = \frac{(273+27)}{(273+327)} = \frac{300}{600} = \frac{1}{2} \Rightarrow V_2 = 2V$.

33. (a) $V \propto T \Rightarrow \frac{V_2}{V_1} = \frac{T_2}{T_1} \Rightarrow \frac{V_2 - V_1}{V_1} = \frac{T_2 - T_1}{T_1}$

$\Rightarrow \frac{\Delta V}{V} = \frac{(273+40)-(273+20)}{(273+20)} = \frac{(313-293)}{293} = 0.07$

34. (d) $P \propto \frac{1}{V} \Rightarrow \frac{V_2}{V_1} = \frac{P_1}{P_2} = \frac{100}{105} \Rightarrow V_2 = \frac{100}{105} V_1 = 0.953\ V_1$

% change in volume $= \frac{V_1 - V_2}{V_1} \times 100$

$= \frac{V_1 - 0.953V_1}{V_1} \times 100 = 4.76\%$

35. (a)

36. (a) At constant temperature

PV = constant

$\Rightarrow \frac{P_1}{P_2} = \frac{V_2}{V_1} \Rightarrow \frac{70}{120} = \frac{V_2}{1200} \Rightarrow V_2 = 700\ ml.$

37. (c) The given equation is for 1 *gm mol* gas.

38. (a) For 1*gm* gas $PV = rT = \left(\frac{R}{M}\right).T$

Since *P* and *V* are constant $\Rightarrow T \propto M \Rightarrow \frac{T_{N_2}}{T_{O_2}} = \frac{M_{N_2}}{M_{O_2}}$

$\Rightarrow \frac{T_{N_2}}{(273+15)} = \frac{28}{32} \Rightarrow T_{N_2} = 252K = -21°C$

39. (a) $PV = \mu RT \Rightarrow PV \propto T$

If *P* and *V* are doubled then *T* becomes four times *i.e.*, $T_2 = 4T_1 = 4 \times 100 = 400K$.

40. (c) At low pressure and high temperature real gases behaves like ideal gases.

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